

Musculoskeletal injuries in rock climbing: a scoping review

Jared Vagy ¹, Alessandra Marenzi,¹ Billy Shiou,¹ Kaikanani Woollings,² Karin Saric³

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ABSTRACT

This scoping review examined the prevalence and characteristics of musculoskeletal injuries in rock climbing and identified existing knowledge and methodology gaps. The methodological guidance of the Joanna Briggs Institute (JBI) was followed. Forty-five studies examining musculoskeletal injuries in rock climbing and climbing-specific training were included, encompassing climbers of various skill levels and age groups. The most common injuries were ligament sprains and muscle or tendon strains, particularly in the fingers, hands, ankles and shoulders. Finger pulley injuries were frequent in both sport climbing and bouldering, while ankle sprains were common in bouldering. Across both acute and chronic categories, finger and ankle injuries were predominant. Future studies with a broader inclusion of under-represented populations (paediatric, women and older adults) and standardised diagnostic reporting are recommended to better understand specific injury mechanisms to reduce injury risk and promote safer climbing practices, as well as to inform clinical practice by supporting the development of targeted interventions.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Rock climbing participation has increased worldwide, alongside a rise in both acute and chronic musculoskeletal injuries, yet reporting remains inconsistent because of variations in methodology, diagnostic detail and population representation.

WHAT THIS STUDY ADDS

⇒ Across 45 studies, the most frequently reported injuries were ligament sprains and muscle or tendon strains, most often involving the fingers, hands, ankles and shoulders.
⇒ Finger pulley injuries were consistently observed in both sport climbing and bouldering, whereas ankle sprains were particularly common in bouldering due to fall-related mechanisms.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ These findings highlight the need for standardised injury reporting and more inclusive research studies related to rock climbing injuries.

INTRODUCTION

Participation in the sport of rock climbing has been steadily rising since its debut in the 2020 Tokyo Olympics.¹ Climbing can be stratified into three main categories: sport climbing, bouldering and speed climbing. Sport climbing uses a rope and harness system, which provides safety during falls. A climber uses either temporary protective equipment (traditional climbing) or permanent anchors as they ascend.^{2 3} Bouldering refers to climbing that is done without a rope and is generally limited to short climbs with foam mats and spotters to provide safety during falls.^{2 3} While sport climbing tends to require more endurance compared with other disciplines, bouldering typically involves fewer, but possibly more technical or powerful movements. Speed climbing is performed on artificial walls where climbers ascend identical routes side by side as quickly as possible while using rope protection.⁴ Climbing disciplines can be further classified by the setting,

either outdoor on natural rock or indoor on artificial rock surfaces.

With the increase in popularity of climbing, an increase in musculoskeletal injuries, both acute and chronic, can be expected.⁵ While this rapid growth is currently being reflected in the literature, there have been no recent reviews published on the topic of musculoskeletal injuries in rock climbing. As such, there is a rising need for the summarisation and synthesis of current data on climbing injuries and risk factors in order to provide up-to-date evidence-based treatments and injury prevention strategies.

The objective of this scoping review was to investigate and summarise the existing evidence on musculoskeletal injuries associated with rock climbing. This review aims to provide a summary of the current data on rock climbing injuries in regard to study characteristics, climbing discipline, climbing setting, mechanism of injury and specific diagnosis. By summarising the available evidence, this review will help to inform future research



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¹Division of Biokinesiology and Physical Therapy, University of Southern California, Los Angeles, California, USA

²Private Practice Physical Therapist, Canmore, Alberta, Canada

³Norris Medical Library, University of Southern California, Los Angeles, California, USA

Correspondence to

Dr Jared Vagy; vagy@usc.edu

directions and strategies for injury prevention and management.

REVIEW QUESTIONS

- What does the current evidence suggest are the prevalence and characteristics of musculoskeletal injuries in rock climbing?
- Which data collection methods were used to obtain the data?
- What are the knowledge or methodology gaps in rock climbing literature related to musculoskeletal injuries?

METHODS

Prior to initiating the review, the protocol was uploaded to Open Science Framework (OSF) and can be found at <https://osf.io/9m87y/>. This scoping review considered full-text articles that focused on musculoskeletal injuries in the context of rock climbing or rock climbing-specific training. Studies in all languages were originally considered, but for feasibility, only articles published in English were included.⁶ Eligible sources included a wide range of study designs, such as self-report surveys, registry data, quantitative and qualitative research methods, retrospective, prospective, cross-sectional and cohort studies, as well as peer-reviewed and non-peer-reviewed studies, white papers and reports. Review articles and case reports were excluded due to concerns about selection bias and limited generalisability. Studies examining non-orthopaedic injuries such as head injuries, concussions, traumatic brain injuries and facial, dental or vascular injuries were excluded. Additionally, research focusing on mountaineering, ice climbing and alpine climbing was not considered for the purposes of this study.

This scoping review was conducted in accordance with the Joanna Briggs Institute (JBI) methodology for scoping reviews and reported using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews (PRISMA-ScR) framework (online supplemental appendix I).^{7,8}

SEARCH STRATEGY

A three-step search strategy was used to locate published studies. A health sciences librarian conducted an initial limited search of MEDLINE (PubMed) and Embase to

identify articles on this topic. The text words contained in the titles and abstracts of relevant articles, and the index terms used to describe the articles were used to develop a full search strategy for MEDLINE (PubMed) (see [table 1](#)).

The health sciences librarian adapted the search strategy, including all identified keywords and index terms, and conducted a second search of five databases on 12 September 2025: MEDLINE (PubMed), Embase, CINAHL Complete (EBSCOhost), Sports Medicine & Education Index (ProQuest) and SPORTDiscus (EBSCOhost) (see online supplemental appendix II). The keywords 'rock climbing' and 'musculoskeletal injury/injuries' were used and adapted as necessary to the scope of the resource. No limits or date restrictions were applied.

Researchers also conducted backward citation searching, searching the reference lists of included studies and any relevant systematic or scoping reviews for additional relevant citations that were not found via database searching. Forward citation searching of included studies was conducted via Google Scholar without filters to identify any citations that cited the included studies. Google Chrome was used in incognito/private mode to mitigate the influence of location and past searches. The first 200 results in Google Scholar were screened for relevance.⁹

Study selection

Following the search, all identified citations were uploaded to Covidence for deduplication and screening. Following a pilot test, where it was determined that the spreadsheet needed to include preset drop-down categories to more easily organise the data, titles and abstracts were screened by two independent reviewers (AM, BS) for assessment against the inclusion criteria for the review.

The full texts of selected citations were assessed in detail against the inclusion criteria by the same two reviewers. Any disagreements that arose between the reviewers at each stage of the selection process were resolved through discussion and a re-review, and finally with a third reviewer (KW). Cohen's Kappa was calculated to assess inter-rater agreement among reviewers for full-text selection. Reasons for exclusion were presented

Table 1 Search strategy for MEDLINE (PubMed)

Concepts	Search terms
Rock climbing	((Rock(tiab)) OR sport(tiab)) OR lead(tiab)) OR route(tiab)) OR speed(tiab)) OR Olympic(tiab)) OR compet*(tiab)) OR wall(tiab)) OR outdoor(tiab)) OR indoor(tiab)) OR "high level"(tiab)) OR climb*(tiab)) OR (Bouldering(tiab))) OR (Top rop*(tiab))
Musculoskeletal injuries	((("Wounds and Injuries"(Mesh)) OR ((climb*(tiab)) OR sport(tiab)) OR athletic(tiab)) OR pulley(tiab)) OR acute(tiab)) OR musculoskeletal(tiab)) OR extremity(tiab)) OR injur*(tiab))) OR (UIAA MedCom Score)

* indicates truncation searching for multiple root endings.

Mesh, Medical Subject Headings; tiab, title/abstract field; UIAA, Union Internationale des Associations d'Alpinisme.

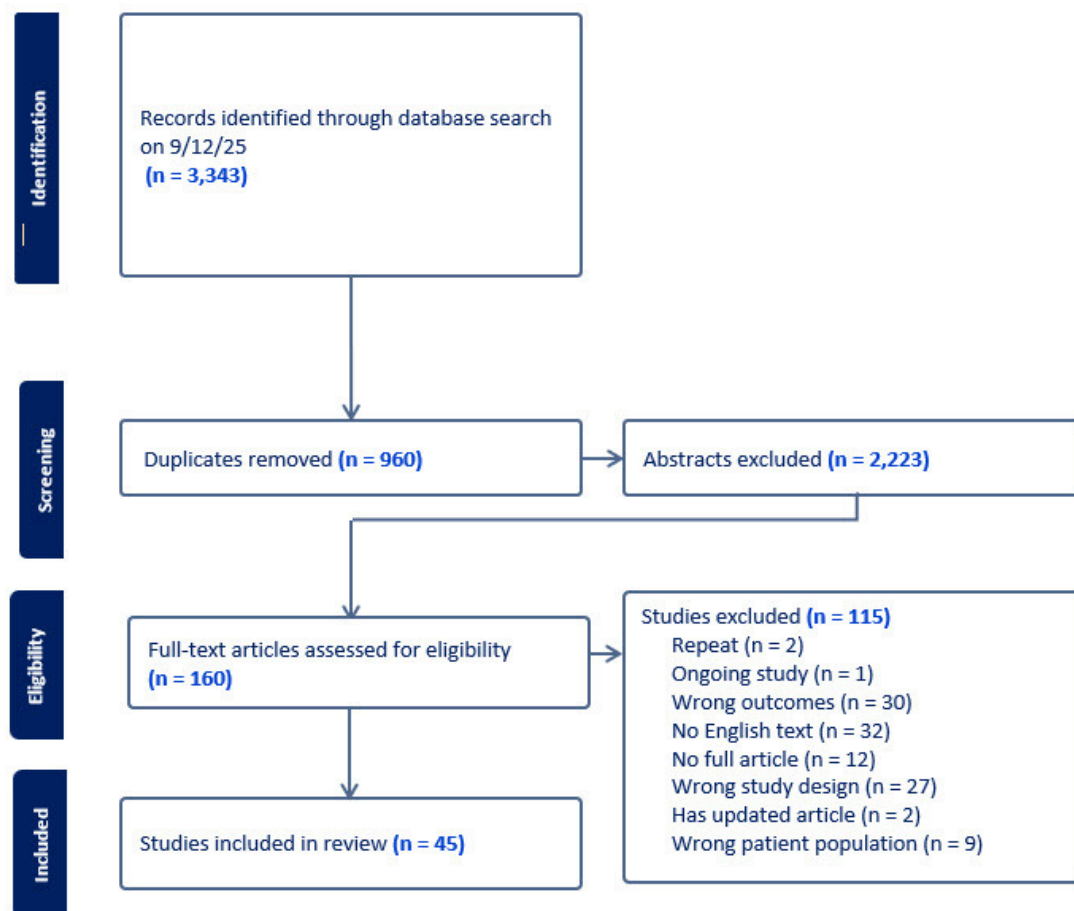


Figure 1 PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) flow diagram for scoping review process.

in the Preferred Reporting Items for Systematic Reviews and Meta-analyses Extension for Scoping Review (PRISMA-ScR) flow diagram (figure 1).⁸

DATA EXTRACTION

Data extraction was conducted by two independent reviewers (AM, BS) using a structured Excel spreadsheet created by the team, with a third reviewer (KW) resolving discrepancies. A shared spreadsheet was used to systematically organise the extracted data, capturing author, year of publication, study design, sample size, geographic region, climbing discipline, climbing setting, mechanism of injury, top five injury locations reported and top 10 injury diagnoses. A draft extraction form is provided (see online supplemental appendix III). The draft data extraction tool was modified and revised after five studies to clarify the data collection methods in the studies and redefine the climbing disciplines.

RESULTS

Forty-five studies were included in this review after initial identification, screening and review of eligibility criteria (figure 1). With regard to study inclusion, Cohen's Kappa was calculated to assess inter-rater reliability for full-text inclusion and was found to be substantial at 0.72.

A portion of the studies included only acute (14/45, 31.1%) or only chronic injuries (4/45, 8.9%), with most studies documenting both (26/45, 57.8%). Data collection methods included cross-sectional (33/45, 73.3%), prospective (5/45, 11.1%) and retrospective (7/45, 15.6%) designs. Sample sizes ranged from <50 participants (7/45, 15.6%) to >500 (11/45, 24.4%). Climbing disciplines examined included bouldering only (4/45, 8.9%), sport climbing only (5/45, 11.1%), both bouldering and sport climbing (27/45, 60.0%) and (4/45, 8.9%), both sport climbing and speed climbing (1/45, 2.2%), and all three disciplines (4/45, 8.9%). Settings included indoor (9/45, 20%), outdoor (3/45, 6.7%) and both (23/45, 51.5%). The reviewed data were collected from six different geographic regions, primarily North America (14/45, 31.3%) and Europe (17/45, 37.8%), with several studies including data which were collected across multiple geographic regions (6/45, 13.3%; figure 2). Studies are summarised in table 2 and online supplemental appendix IV.

The populations in the reviewed studies include athletes of various levels, ranging from recreational to competitive, and across age groups from paediatric to older adults. The selected studies were published between 1990 and 2025. Fifteen studies included clinician-confirmed

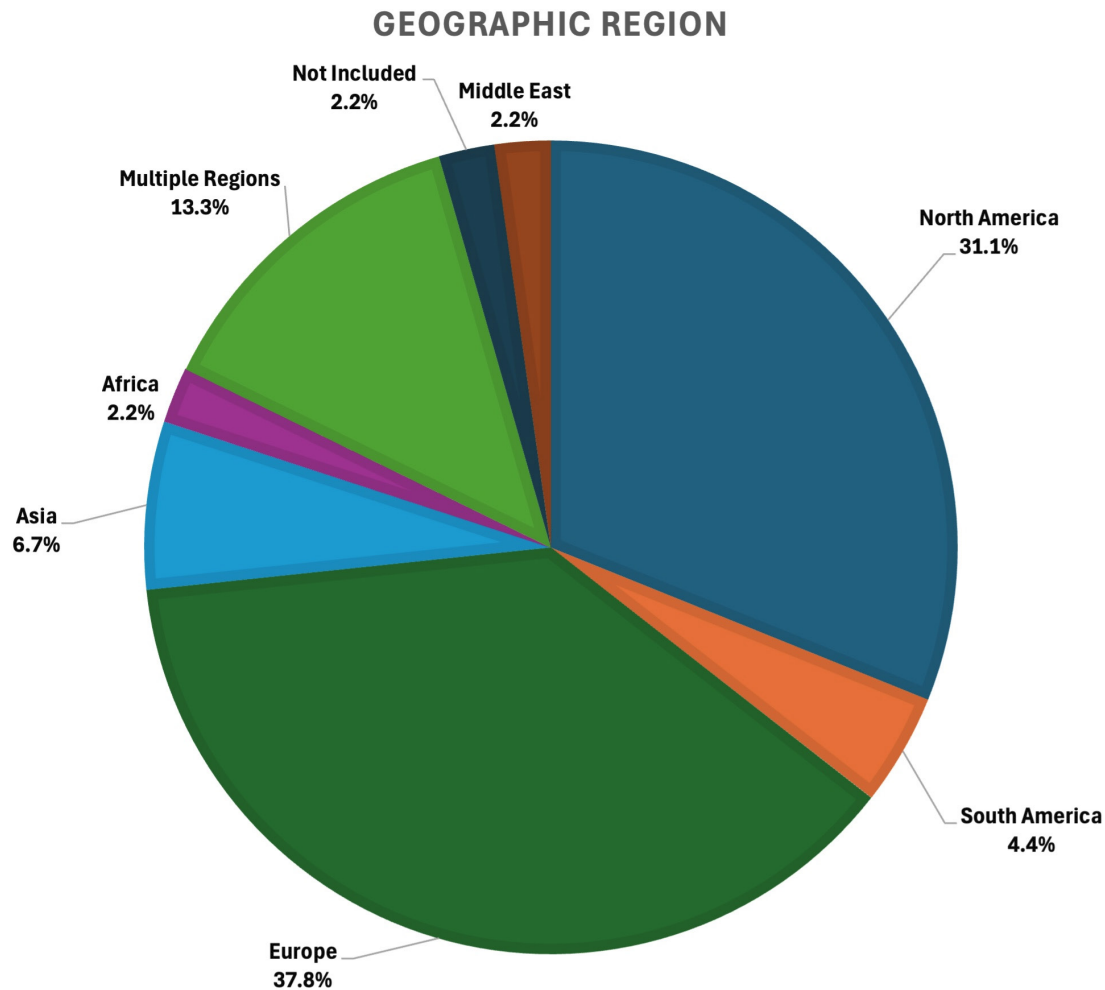


Figure 2 Geographic regions of study.

diagnosis, while 30 studies obtained data through self-report methods only. Six studies reported musculoskeletal injuries to a single body region including hand, wrist/hand, wrist, upper extremity in general and knee.^{7,10–14} Of the selected studies, 31 provided information regarding specific injury diagnoses, both self-reported and diagnosed by a healthcare provider.

Injury diagnoses

Nineteen studies included information regarding injury diagnoses for both acute and chronic injuries, 12 studies for acute injuries only and 1 study reported diagnoses for chronic injuries only. Of these, 21 found that ligament sprains and muscle or tendon strains were the most frequent acute injuries in rock climbing, with many identifying specific location or structure (ie, finger tendons, A2 pulley, ankle ligaments, etc). In contrast, four studies investigating acute injury found fractures to be the most common,^{15–18} although three of these studies sampled from hospitals only.^{15,17,18} With regard to lower extremity injuries, four studies spanning all three disciplines found that ankle sprains were the most common acute injury in climbing,^{15,19–21} with an additional study listing ‘soft tissue ankle injury’ as most prevalent.²²

A single study examining chronic injury in the hands and elbows found that proximal interphalangeal joint collateral ligament sprains, occurring in 40.5% of elite climbers sampled, were most prevalent.¹³

With regard to injury location, the hand/finger was the most commonly reported area based on studies that reported injuries from all body regions (figure 3), again with ligament sprains and muscle or tendon strains as the most prevalent diagnoses in these locations. Pulley injuries were further specified as the top upper extremity diagnosis in six studies.^{10,23–27}

Injuries by climbing discipline

For studies that reported data on bouldering alone, the most common injury locations were the finger, hand and ankle.^{21,28–30} The ankle was the most frequent injury location due to falls in both indoor and outdoor bouldering, although the prevalence of ankle injury was higher in outdoor bouldering.²⁹ Additionally, ankle injuries were the most common reason for emergency department visits resulting from indoor bouldering.²¹ Overall, the hand and fingers continue to be the most common injury location considering both discipline and setting, although there may be evidence of a higher prevalence

Table 2 Study characteristics

Variable	Frequency (%)
Data collection method	
Cross sectional	33 (73.3%)
Prospective	5 (11.1%)
Retrospective	7 (15.6%)
Sample size	
<50	7 (15.6%)
50–100	7 (15.6%)
101–200	8 (17.8%)
201–300	6 (13.3%)
301–400	1 (2.2%)
401–500	5 (11.1%)
>500	11 (24.4%)
Geographic region	
North America	14 (31.1%)
South America	2 (4.4%)
Europe	17 (37.8%)
Asia	3 (6.7%)
Africa	1 (2.2%)
Middle East	1 (2.2%)
Multiple regions	6 (13.3%)
Missing/not included	1 (2.2%)
Climbing discipline	
Bouldering	4 (8.9%)
Sport climbing	5 (11.1%)
Speed climbing	0 (0%)
Bouldering and sport climbing	27 (60.0%)
Sport climbing and speed climbing	1 (2.2%)
All three disciplines	4 (8.9%)
Missing/not included	4 (8.9%)
Climbing setting	
Indoor	9 (20.0%)
Outdoor	3 (6.7%)
Both	23 (51.1%)
Missing/not included	10 (22.2%)
Mechanism of injury	
Acute	14 (31.1%)
Chronic	4 (8.9%)
Both	26 (57.8%)
Not stated	1 (2.2%)

of acute finger injuries occurring in outdoor settings compared with indoor.²⁹

Five studies examined injury in sport climbing alone.^{10 13 27 31 32} One study investigated nerve compression syndromes exclusively, reporting that approximately 25% of sport climbers presenting with complaints of

the upper or lower extremity had a nerve compression syndrome.³² Another study examined hand injuries only, finding A2 pulley injury of the fourth finger to be the most common.¹⁰ Of the remaining three studies, all found finger ligament injuries to be most prevalent.^{13 27 31} Ultimately, pulley and ligament injuries in the fingers were the top reported injuries in studies examining sport climbing. Injury definitions throughout these studies were heterogeneous, making comparison difficult as some reported time-loss injuries, others only acute injuries, while others reported all discomforts including those requiring no time loss from sport.

Five of the studies selected included speed climbing in addition to bouldering and/or sport climbing.^{20 33–36} All examined competitive climbers, three focusing on youth^{20 33 36} and one on females exclusively.³⁴ One found the most commonly reported diagnosis among competitive youth climbers across all disciplines to be primary periphyseal stress injury to the fingers, followed by finger capsulitis,³⁶ while another reported the top injury to be ankle sprain, followed by pulley injury, although this study used self-reported findings not diagnosed by a healthcare practitioner.²⁰ Two others which reported youth climbing injury by anatomical location reported hands/fingers to be the most common locations.³³

DISCUSSION

Of the studies reviewed, the majority focused on samples from North America and Europe, making these findings most applicable to climbers in these continents. However, the populations encompassed various levels and age groups, making these results broadly generalisable to a wide range of climbers.

The finding that the hand and fingers are the most common sites of both acute and chronic injuries aligns with existing literature,^{10 11 13 23–30 33 37–50} reinforcing the understanding that rock climbing places substantial stress on these areas due to the high demands for grip strength and finger loading. While rehabilitation and injury prevention strategies from other sports can offer valuable insights, climbing-specific programmes are crucial for effectively preparing climbers to safely return to their sport. Tailoring interventions to the unique physical demands of climbing can help address the sport's specific risks and enhance recovery outcomes.

Ligament sprains and muscle or tendon strains were the most frequently reported injuries for both acute and chronic cases.^{2 10 11 13 19–21 23–27 31 38 40 46–49 51 52} However, studies conducted in emergency room and surgery department settings commonly reported fractures, likely due to the fact that only severe, acute injuries typically require emergency and surgical care, while less severe sprains and strains often go unreported in such settings. This discrepancy underscores the potential underrepresentation of milder injuries in hospital data. Additionally, less severe injuries that do not prevent climbers from continuing the sport (ie, those resulting in no time lost

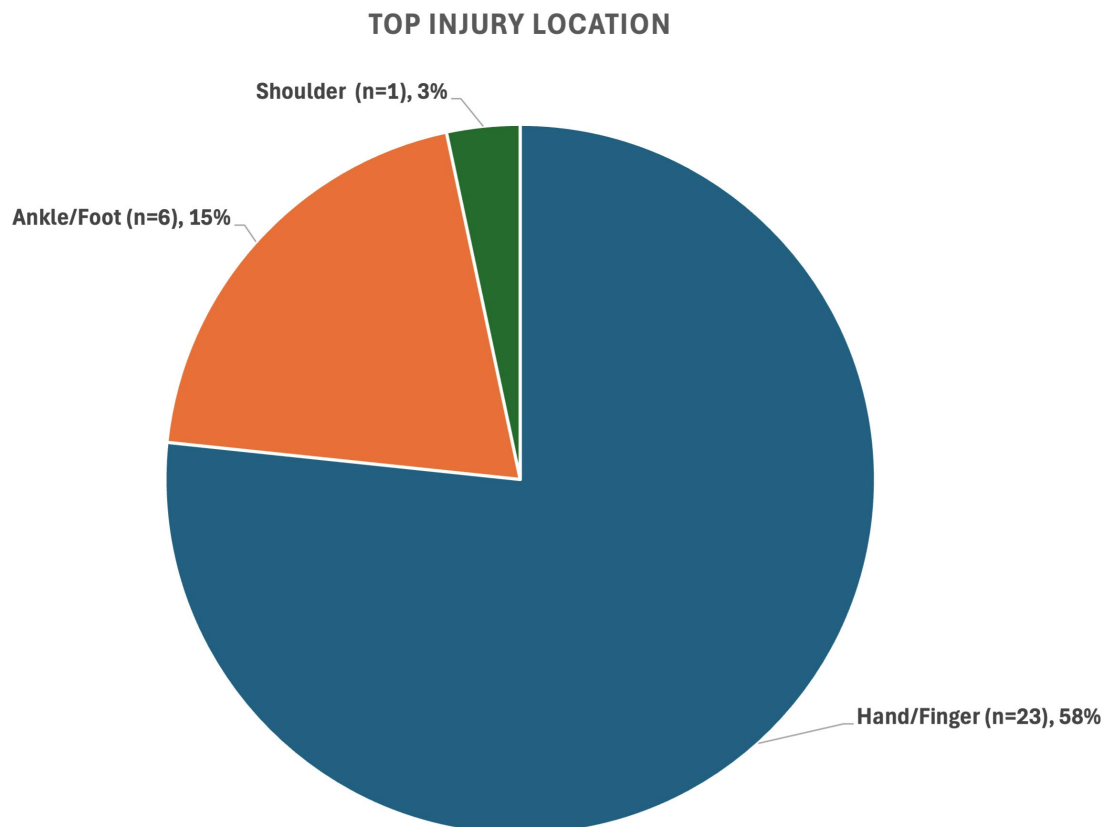


Figure 3 Top injury location.

from climbing) may go unreported both in hospital settings and as well as other study contexts.

In studies examining sport climbing alone, finger ligament and pulley injuries were the most commonly reported.^{10 13 27 32} Again, this can be explained by the high loads and volume that climbers exert through the fingers. A greater body of evidence is needed to make firm conclusions about finger injuries in the discipline of bouldering; however, some studies support similar conclusions as in sport climbing.^{28 29} In studies that included speed climbers, diagnoses were only provided in two.^{35 53} Ankle injuries, both fractures and sprains, were common, as well as injuries to the fingers.^{35 36} A focused analysis of speed climbing alone was not possible as all speed climbers in the reviewed studies also participated in other climbing disciplines. Consequently, identifying injury patterns and risk factors specific to speed climbing is challenging, highlighting the need for further research, particularly randomised controlled trials that can control for confounding variables. However, the fact that many speed climbers cross-train in other climbing styles or sports enhances the generalisability of the findings to diverse climbing populations.

Although upper extremity injuries far outweighed any others, lower extremity injuries were still common. The ankle was reported to be the most commonly injured area in the lower extremity.^{19 21 22 34 42 52 53} This may be due to the fact that falling from a route or boulder problem is common, often leading to trauma when hitting the wall

or ground. It is also possible that climbing shoe design and wear may lead to injury in the foot and ankle region; however, further research is needed on this topic. Several studies also reported knee injuries to be moderately common in climbing,^{2 21 34 51 54} which corroborates more recent research showing a rise in acute knee injuries with the growth of indoor bouldering and mechanisms such as heel hooks, falls, high steps and drop knees.¹²

Pulley ruptures and finger tendon strains were the predominant upper extremity diagnoses, reflecting overload from repetitive high-force gripping. Management typically involves temporary load reduction, progressive reloading and targeted rehabilitation. In the lower extremity, ankle sprains and fractures were most common, usually from falls. Rehabilitation in these cases would typically be focused on restoring mobility, proprioception, strength and dynamic stability.

In this review, more studies collected data from indoor settings than outdoors, although many articles studied both settings. This is likely due to ease of data collection, as well as the higher volume of climbers visiting indoor gyms. This review highlights the need for more research into the climbing setting as a risk factor for injury, and whether the setting affects the type and severity of injury. It can be speculated that more severe injuries occur in outdoor climbing compared with indoor climbing due to the greater safety measures typically enforced in climbing gyms and the lack of prerequisites for climbing outdoors. Additionally, outdoor environments may present

challenges such as limited access to help and unpredictable factors like terrain and weather, all of which can increase the risk and severity of injuries.

The reporting of specific diagnoses was limited, and when included, the level of diagnostic detail was often insufficient. As mentioned, many studies used self-reported injury data, making specific diagnoses unclear. This lack of specificity limits the ability to accurately identify injury trends and patterns, which is essential for developing effective preventive measures and treatment protocols.

There was also a lack of demographic diversity in the study populations. Key groups such as paediatric, older adults, women and non-binary gender groups appeared to be underrepresented. Seven studies (15.6%) included youth climbers, three studies sampled over 50% percent women, and no studies identified non-binary gender groups. However, recent studies have begun to show a trend towards inclusion of both women and paediatric climbers. Ethnicity was not included in reporting, limiting the generalisability of the findings across different climber demographics.

This review included a variety of study designs, the majority of which derived from self-reported, cross-sectional studies, making results susceptible to recall bias, self-reporting errors and selection bias. With only five studies providing prospective insights, the lack of longitudinal studies also limits the ability to understand how injuries develop and persist over time, and the nature of temporal relationships and causation.

It was outside the scope of this review to appraise the quality of each study included, limiting the ability to properly weight higher quality evidence above lower quality, as well as making it impossible to calculate statistics and synthesise data. As more studies with less heterogeneity in study design become available, meta-analyses can be conducted. Overall, this broad approach provided a comprehensive view of the existing research and helped identify gaps in the current knowledge. Future studies with more detailed and comprehensive diagnostic reporting would improve our understanding of specific injury mechanisms and aid in creating targeted interventions to reduce risk and promote safer climbing practices.

Limitations

Several factors restricted the capacity to draw reliable conclusions from the data available for this review. The findings are limited by the reliance on self-reported, cross-sectional data, which introduces recall and selection bias. Few longitudinal studies exist, reducing insight into injury progression and causation. Significant heterogeneity in study design further complicates comparison and makes statistical analysis and therefore synthesis of incidence and prevalence values impossible. For feasibility, non-English literature was excluded. The majority of non-English literature was obtained as scanned PDFs through library reserves, and it was not feasible to translate without a native speaker of the language. This

limited the available data sources from non-English-speaking countries. The research also lacked geographic diversity. Regions with growing climbing communities, such as Asia and South America, were underrepresented in the literature. This gap could lead to a skewed understanding of injury patterns that may be unique to these regions due to differences in climbing styles, training techniques and environmental conditions. While this review highlights the nature of musculoskeletal injuries in rock climbing, these limitations point to substantial areas for improvement in future research, particularly in enhancing methodological rigour and increasing the diversity of study populations.

CONCLUSIONS

The inconsistencies in injury rates, study populations, settings and climbing disciplines across current research underscore the need for standardised injury reporting and clearer categorisation of these variables. Consistent with previous studies, the hands and fingers were the most commonly injured areas. Both acute and chronic mechanisms frequently caused ligament sprains and muscle or tendon strains, due to the high loads and repetitive strain climbers placed on the fingers. Although less common, lower extremity injuries also occurred, particularly to the ankle and knee, often as a result of falls. Clinically, the predominance of pulley and tendon injuries in the fingers and sprains or fractures in the ankle highlights the need for focused prevention and rehabilitation strategies targeting these tissues. Future prospective studies targeting hand and finger injuries, as well as ankle injuries, will be needed to help enhance our understanding of injury mechanisms, risk factors involved and to establish prevalence and incidence rates. These will help to inform targeted interventions that effectively reduce injury risk and support safer climbing practices. With the sport's growing popularity, establishing preventive strategies and educating climbers and healthcare providers on common injury types, mechanisms and risk factors will be essential.

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ORCID iD

Jared Vagy <https://orcid.org/0009-0007-8589-1138>

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